

A dichotomy for transitive lists

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Joint work with Roy Shalev and Stevo Todorčević

Cellularity of the real line

It is easy to see that the real line is ccc - it does not contain an uncountable family of pairwise disjoint open sets.

This actually holds for any separable linear order.

Souslin's problem - cellularity vs density

Fundamenta Mathematicae 1, 1920:

3) Un ensemble ordonné (linéairement) sans sauts ni lacunes et tel que tout ensemble de ses intervalles (contenant plus qu'un élément) n'empiétant pas les uns sur les autres est au plus dénombrable, est-il nécessairement un continu linéaire (ordinaire)?

Problème de M. M. Souslin.

Souslin's problem

If a linear order is complete, dense, without endpoints, and ccc, is it necessarily isomorphic to the real line (separable)?

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Reformulation of Souslin's problem

Kurepa observed in 1930s that Souslin's problem is in fact (equivalent to) a problem about trees.

Trees

An ordinal is a transitive set well-ordered by $\in$.

Equivalently, an ordinal is the order type of a well-ordered set.

A tree is a partially ordered set $(T, <_T)$ such that for every $t \in T$, the set of its predecessors $t_{\downarrow} = \{x \in T : x <_T t\}$ is well-ordered.

We define the α -th level of T as $T_{\alpha} = \{t \in T : \text{otp}(t_{\downarrow}, <_T) = \alpha\}$.

Then the height of T is $\text{ht}(T) = \min \{\alpha : T_{\alpha} = \emptyset\}$.

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κ -Trees

We say that $(T, <_T)$ is a κ -tree if

- $(t_\downarrow, <_T)$ is well-ordered for every $t \in T$;
- $|T_\alpha| < \kappa$ for each ordinal α ;
- $\text{ht}(T) = \kappa$.

If $(\kappa, <_T)$ is an ordinal-based κ -tree, i.e. $\alpha <_T \beta \Rightarrow \alpha < \beta$, then:

- $(\forall \delta < \kappa) \delta_\downarrow \subseteq \delta$;
- $(\forall \alpha < \kappa) |\{\delta < \kappa : \text{otp}(\delta_\downarrow, <_T) = \alpha\}| < \kappa$;
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Souslin trees

We say that a tree T is *Souslin* if it is of size $\aleph_1$ and contains neither chains nor antichains of size $\aleph_1$.

Theorem (Kurepa, 1930s)

The answer to Souslin's problem is positive iff there are no Souslin trees.

A tree T is *Aronszajn* if it is an ω_1 -tree and contains no chains of size $\aleph_1$.

Theorem (Aronszajn, 1930s)

There is an Aronszajn tree.

Theorem (Solovay-Tennenbaum, 1971)

Consistently, there are no Souslin trees.

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Martin's axiom $\text{MA}_{\aleph_1}$ - Martin and Solovay, late 1960s

For every ccc poset $\mathbb{P}$ and every family $\{D_\alpha : \alpha < \omega_1\}$ of dense subsets of $\mathbb{P}$, there is a filter $G \subseteq \mathbb{P}$ such that $G \cap D_\alpha \neq \emptyset$ for every $\alpha < \omega_1$.

Theorem (Baumgartner-Malitz-Reinhardt, 1970)

$\text{MA}_{\aleph_1}$ implies that every Aronszajn tree can be decomposed into countably many antichains, i.e. every Aronszajn tree is special.

In particular, $\text{MA}_{\aleph_1}$ implies that there are no Souslin trees.

Problem

Direct generalization of Martin's axiom to $\aleph_2$ -cc posets is not possible.

Theorem (Laver-Shelah, 1980)

It is consistent, assuming the existence of a weakly compact cardinal, that the Continuum Hypothesis holds and there are no $\aleph_2$ -Souslin trees.

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For the remainder of the talk $\lambda < \kappa$ are regular infinite cardinals.

Trees, repeated

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Transitive lists

Definition

A sequence $\vec{L} := \langle L_\delta \mid \delta \in \Delta \rangle$ is a κ -list with support Δ if $\Delta \in [\kappa]^\kappa$ and

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If a κ -list $\vec{L}$ additionally satisfies

- for every $\gamma < \delta$ in Δ , if $\gamma \in L_\delta$, then $L_\gamma \subseteq L_\delta$,

then we say that $\vec{L}$ is a *transitive* κ -list.

For $\alpha < \kappa$ denote $\text{Lev}_\alpha(\vec{L}) = \{\delta \in \Delta \mid \text{otp}(L_\delta) = \alpha\}$.

We denote by $\text{TL}(\kappa)$ the class of all transitive κ -lists.

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Trees and transitive lists

For a transitive κ -list $\vec{L} = \langle L_\delta : \delta < \kappa \rangle$ define the relation $\prec_{\vec{L}}$ on κ by

$$\gamma \prec_{\vec{L}} \delta \text{ iff } \gamma \in L_\delta.$$

Lemma

If $\vec{L}$ is a transitive κ -list, then $(\kappa, \prec_{\vec{L}})$ is a tree if and only if $\langle L_\gamma : \gamma \in L_\delta \rangle$ is injective for each $\delta < \kappa$.

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Transitive lists, cont'd

Given $\vec{L} := \langle L_\delta \mid \delta \in \Delta \rangle$ and $A \subseteq \Delta$, we define $\vec{L} \restriction A := \langle L_\delta \mid \delta \in \Delta \cap A \rangle$.

For $X \subseteq \kappa$, we measure the minimal size of a family needed to cover X :

$$\rho_{\vec{L}}(X) := \min\{\theta \leq \kappa \mid \kappa = \theta \text{ or } (\exists F \in [\Delta]^\theta) X \subseteq \bigcup_{\beta \in F} L_\beta\}.$$

Lemma

$\vec{L} = \langle L_\delta : \delta \in \Delta \rangle$ belongs to $\text{TL}(\kappa)$ iff it satisfies the following conditions:

- (1) $\Delta \in [\kappa]^\kappa$;
- (2) $L_\delta \subseteq \delta$ for every $\delta \in \Delta$;
- (3') $C(\vec{L}) := \{\gamma \in \text{acc}(\kappa) \mid (\forall \delta \in \Delta \setminus \gamma) \sup(L_\delta \cap \gamma) = \gamma\} \in [\kappa]^\kappa$;
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The list dichotomy

Definition

For cardinals θ_0, θ_1 , denote by $\text{TL}(\lambda^+) \rightarrow \left(\begin{smallmatrix} \lambda^+ \\ \theta_0 \end{smallmatrix} ; \begin{smallmatrix} \lambda\text{-partition} \\ \theta_1 \end{smallmatrix} \right)$ the assertion that for every transitive λ^+ -list $\vec{L} := \langle L_\delta \mid \delta \in \Delta \rangle$ either:

- there is $A \in [\lambda^+]^{\lambda^+}$ such that $\rho_{\vec{L}}(A \cap \delta) < \theta_0$ for every $\delta < \lambda^+$, or
- a partition $\bigcup_{i < \lambda} A_i = \Delta$ so that for $i < \lambda$ and $X \in [A_i]^{\theta_1}$: $\rho_{\vec{L}}(X) \geq \theta_1$.

Let $\text{LD}(\lambda^+)$ denote $\text{TL}(\lambda^+) \rightarrow \left(\begin{smallmatrix} \lambda^+ \\ 2 \end{smallmatrix} ; \begin{smallmatrix} \lambda\text{-partition} \\ \lambda \end{smallmatrix} \right)$.

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The list dichotomy: slightly reformulated

$\text{LD}(\lambda^+)$:

For every $\vec{L} \in \text{TL}(\lambda^+)$ either

- $(\exists A \in [\lambda^+]^{\lambda^+})(\forall \delta < \lambda^+)(\exists \gamma \in \Delta) A \cap \delta \subseteq L_\gamma$, or
- we can partition $\Delta = \bigcup_{i < \lambda} A_i$ so that

$$(\forall i < \lambda)(\forall X \in [A_i]^\lambda)(\forall \Sigma \in [\Delta]^{<\lambda}) X \not\subseteq \bigcup_{\delta \in \Sigma} L_\delta.$$

Observation

If T is an ordinal-based λ^+ -Aronszajn tree, then the naturally associated $\vec{L} = \langle \delta_\downarrow : \delta < \lambda^+ \rangle$ never satisfies the first alternative of $\text{LD}(\lambda^+)$.

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- a partition $\bigcup_{i < \lambda} A_i = \Delta$ so that $\rho_{\vec{L} \upharpoonright A_i}(X) \geq \theta_1$ for $i < \lambda$, $X \in [A_i]^{\theta_1}$.

And let $\text{LD}_{part,loc}(\lambda^+)$ denote $\text{TL}(\lambda^+) \rightarrow \left(\begin{smallmatrix} \lambda^+ \\ 2 \end{smallmatrix} ; \begin{smallmatrix} \lambda\text{-partition, local} \\ \lambda \end{smallmatrix} \right)$.

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- there is $A \in [\lambda^+]^{\lambda^+}$ such that $\rho_{\vec{L}}(A \cap \delta) < \theta_0$ for every $\delta < \lambda^+$, or
- a partition $\bigcup_{i < \lambda} A_i = \Delta$ so that $\rho_{\vec{L} \upharpoonright A_i}(X) \geq \theta_1$ for $i < \lambda$, $X \in [A_i]^{\theta_1}$.

And let $\text{LD}_{part,loc}(\lambda^+)$ denote $\text{TL}(\lambda^+) \rightarrow \left(\begin{smallmatrix} \lambda^+ \\ 2 \end{smallmatrix} ; \begin{smallmatrix} \lambda\text{-partition, local} \\ \lambda \end{smallmatrix} \right)$.

Dichotomy for many lists

Definition

For λ an infinite regular cardinal, let $\text{LD}(\lambda^+, \lambda)$ be the assertion that for all λ -many lists $\langle \vec{L}_j := \langle L_{\delta,j} \mid \delta \in \Delta_j \rangle \mid j < \lambda \rangle$ in $\text{TL}(\lambda^+)$ either:

- there exists $A \in [\lambda^+]^{\lambda^+}$ such that for every $\delta < \lambda^+$ there exists $j < \lambda$ with $\rho_{\vec{L}_j}(A \cap \delta) = 1$, or
- there exists a sequence $\langle A_i \mid i < \lambda \rangle$ such that $\bigcup_{i < \lambda} A_i = \bigcup_{j < \lambda} \Delta_j$ and, for all $i, j < \lambda$ and $X \in [A_i \cap \Delta_j]^\lambda$ we have $\rho_{\vec{L}_j}(X) \geq \lambda$.

Then $\text{LD}(\lambda^+, \lambda) \Rightarrow \text{LD}(\lambda^+) \Rightarrow \text{LD}_{\text{part}, \text{loc}}(\lambda^+)$.

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Application to trees

Lemma

Let $\vec{L} = \langle L_\delta \mid \delta \in \Delta \rangle$ be a transitive λ^+ -list satisfying the second alternative of $\text{LD}_{part,loc}(\lambda^+)$. Then, the list is special, i.e. there exists a map $c : \Delta \rightarrow \lambda$ such that $\gamma \notin L_\delta$ for all $\gamma, \delta \in \Delta$ satisfying $c(\gamma) = c(\delta)$.

Corollary

*Assume $\text{LD}(\lambda^+)$. Then every λ^+ -Aronszajn tree is special.
In particular λ^+ -Souslin's hypothesis holds under $\text{LD}(\lambda^+)$.*

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Application to lattices

Definition

A lower semi-lattice P is called λ^+ -Aronszajn if it is well-founded and

- P is of size λ^+ ;
- P_α is of size at most λ for all $\alpha < \lambda^+$;
- P does not contain any chain of size λ^+ .

A lower semi-lattice is called λ^+ -Souslin if it is λ^+ -Aronszajn and every set of pairwise incomparable elements is of size $< \lambda^+$.

Theorem (Raghavan – Yorioka, 2014)

- $\text{MA}_{\aleph_1}$ implies there are no Souslin lower semi-lattices.
- $\text{PID} + \mathfrak{p} > \omega_1$ imply there are no Souslin lower semi-lattices.
- It is consistent that there is a Souslin lower semi-lattice and every Aronszajn tree is special.

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Application to lattices, cont'd

A λ^+ -Aronszajn lower semi-lattice $(P, <)$ is called *special* if it can be decomposed into λ -many sets of pairwise incomparable elements, i.e. if there is a map $c : P \rightarrow \lambda$ such that $c^{-1}[\{i\}]$ is a set of pairwise incomparable elements for all $i < \lambda$.

Theorem

Suppose $\text{LD}(\lambda^+)$ holds. Then there are no λ^+ -Souslin lower semi-lattices. Moreover, every λ^+ -Aronszajn lower semi-lattice is special.

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Application to strongly unbounded colorings

Definition (Rinot – Shalev – Todorčević, 2024)

We say that a map $c : \lambda \times \lambda^+ \rightarrow \lambda$ is a *strongly unbounded coloring* if for each $B \in [\lambda^+]^{\lambda^+}$, there are $\eta < \lambda$, $t : \eta \rightarrow \lambda$ such that

$$\sup\{c(\eta, \beta) \mid \beta \in B \text{ \& } t \subseteq c(\cdot, \beta)\} = \lambda.$$

Theorem

Under $\text{LD}(\lambda^+)$, there is no strongly unbounded coloring $c : \lambda \times \lambda^+ \rightarrow \lambda$ such that $\langle c(\cdot, \beta) \mid \beta < \lambda^+ \rangle$ is $\leq^$ -increasing sequence of increasing maps.*

Corollary

Assuming $\text{LD}(\omega_1)$, we have $\mathfrak{b} > \aleph_1$ (this holds under $\text{MA}_{\aleph_1}$ as well).

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A sequence $\mathcal{T} = \{T_\alpha \mid \alpha < \lambda^+\} \subseteq [\lambda]^\lambda$ is a λ^+ -tower in $(\mathcal{P}(\lambda), \subseteq^*)$ if:

$$T_\alpha \subseteq^* T_\beta \text{ iff } \alpha < \beta.$$

A tower $\mathcal{T}$ is *Souslin* if it contains no $\subseteq$ -antichain of cardinality λ^+ ,

A tower $\mathcal{T}$ is *Hausdorff* if there is a map $c : \lambda^+ \rightarrow \lambda$ such that for $i, j < \lambda$ and $\alpha \in c^{-1}[\{i\}]$, the set $\{\xi \in c^{-1}[\{i\}] \cap \alpha \mid T_\xi \setminus T_\alpha \subseteq j\}$ is of size $< \lambda$.

Theorem (Borodulin-Nadzieja – Chodounský, 2015)

Assuming $\text{MA}_{\aleph_1}$, every ω_1 -tower is Hausdorff, in particular non-Souslin.

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Definition

A sequence $\mathcal{G} = \langle (a_\alpha, b_\alpha) \mid \alpha < \lambda^+ \rangle$ of pairs of elements from $[\lambda]^\lambda$ is a (λ^+, λ^+) -pregap in $\mathcal{P}(\lambda)$ if:

- $a_\alpha \subseteq^* a_\beta$ and $b_\alpha \subseteq^* b_\beta$ for all $\alpha < \beta < \lambda^+$,
- $|a_\alpha \cap b_\beta| < \lambda$ for all $\alpha \neq \beta$ in λ^+ ,
- $a_\alpha \cap b_\alpha = \emptyset$ for every $\alpha < \lambda^+$.

$\mathcal{G}$ is a *gap* if there is no $c \in [\lambda]^\lambda$ such that $a_\alpha \subseteq^* c$ and $|b_\alpha \cap c| < \lambda$ for all $\alpha < \lambda^+$.

Recall that a pregap $\mathcal{G}$ is a gap iff for every $I \in [\lambda^+]^{\lambda^+}$, there are α and β in I so that $(a_\alpha \cap b_\beta) \cup (a_\beta \cap b_\alpha) \neq \emptyset$.

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Application to gaps, cont'd

A gap $\mathcal{G}$ is an S -gap iff for every $I \in [\lambda^+]^{\lambda^+}$ there are two distinct ordinals α and β in I such that $(a_\alpha \cap b_\beta) \cup (a_\beta \cap b_\alpha) = \emptyset$;

A gap $\mathcal{G}$ is a T -gap iff for every $I \in [\lambda^+]^{\lambda^+}$ there are two distinct ordinals α and β in I such that $a_\alpha \subseteq a_\beta$ and $b_\alpha \subseteq b_\beta$.

Clearly, any T -gap is an S -gap.

Theorem (Lopez – Todorčević, 2017)

It is consistent that there is an S -gap and that there are no T -gaps.

Theorem (Kunen, 1970s)

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Theorem

Assuming $\text{LD}(\lambda^+)$ there are no (λ^+, λ^+) - T -gaps in $\mathcal{P}(\lambda)$.

Theorem (Shalev–Todorćević)

Suppose that κ is a weakly compact cardinal and $2^\kappa = \kappa^+$. Then there is a partial order $\mathbb{P}$ such that in the forcing extension by $\mathbb{P}$ we have that CH holds and every (ω_2, ω_2) -gap in $(\mathcal{P}(\omega_1), \subseteq^)$ is Hausdorff.*

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Application to Tukey order of directed sets

A directed set $D = (\lambda^+, \leq_D)$ is λ^+ -nice if it is ordinal-based and for every subset X of D of size λ^+ there is an unbounded $Y \subseteq X$ of cardinality λ .

Let D be a λ^+ -nice directed set. We define a forcing notion $\mathbb{A}_D$ of conditions of the form $q = (A_q, B_q)$ where $A_q, B_q \in [\lambda^+]^{<\lambda}$.

The order $\leq_{\mathbb{A}_D}$ is defined by $q_1 \leq_{\mathbb{A}_D} q_0$ iff:

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Lemma

$\mathbb{A}_D$ is $<\lambda$ -closed, and adds $Y \subseteq \lambda^+$ of size $(\lambda^+)^V$ so that every λ -sized subset of Y is $\leq_D$ -unbounded. In particular $D \equiv_T [(\lambda^+)^V]^{<\lambda}$ is forced.

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Let D be a λ^+ -nice directed set. We define a forcing notion $\mathbb{A}_D$ of conditions of the form $q = (A_q, B_q)$ where $A_q, B_q \in [\lambda^+]^{<\lambda}$.

The order $\leq_{\mathbb{A}_D}$ is defined by $q_1 \leq_{\mathbb{A}_D} q_0$ iff:

- $A_{q_1} \supseteq A_{q_0}$,
- $B_{q_1} \supseteq B_{q_0}$,
- $\alpha \not\leq_D \beta$ for every $\alpha \in A_{q_1} \setminus A_{q_0}$ and $\beta \in B_{q_0}$.

Lemma

$\mathbb{A}_D$ is $<\lambda$ -closed, and adds $Y \subseteq \lambda^+$ of size $(\lambda^+)^V$ so that every λ -sized subset of Y is $\leq_D$ -unbounded. In particular $D \equiv_T [(\lambda^+)^V]^{<\lambda}$ is forced.

Application to Tukey theory of directed sets, cont'd

Theorem

Suppose $\text{LD}(\lambda^+)$ holds and that $\lambda^{<\lambda} = \lambda$. If $D <_T [\lambda^+]^{<\lambda}$ is a directed set such that every $X \subseteq D$ of size λ^+ contains an unbounded subset of size λ , then there is a λ^+ -c.c. forcing so that in the extension $D \equiv_T [\lambda^+]^{<\lambda}$.

Corollary (Todorćević, 1985)

Under $\text{MA}_{\aleph_1}$, if D is a directed set of size $\aleph_1$ whose each uncountable subset contains an infinite unbounded subset, then $D \equiv_T [\omega_1]^{<\omega}$.

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Consistency at $\aleph_1$

Definition (Chodounský – Zapletal, 2015)

A poset P satisfies Y -c.c. if for every countable elementary submodel $M \prec H_\theta$ containing P , and every condition $q \in P$, there is a filter $F \in M$ on the completion $\text{ro}(P)$ such that $\{p \in \text{ro}(P) \cap M : p \geq q\} \subseteq F$.

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Theorem (Chodounský – Zapletal, 2015)

Every Y -c.c. poset is c.c.c.

Consistency at ω_1 , cont'd

Definition

P is *strong* Y -c.c. if there is a function w defined on P such that

- for every $p \in P$, $w(p)$ is a finite set;
- if $p, q \in P$ are compatible, then they have a lower bound $r \leq p, q$ such that $w(r) = w(p) \cup w(q)$;
- whenever $\langle p_\alpha \mid \alpha < \omega_1 \rangle$ and $\langle q_\alpha \mid \alpha < \omega_1 \rangle$ are subsets of P such that $\langle w(p_\alpha) \mid \alpha < \omega_1 \rangle$ and $\langle w(q_\alpha) \mid \alpha < \omega_1 \rangle$ are Δ -systems with the same root, there are ordinals $\alpha, \beta < \omega_1$ so that p_α and q_β are compatible.

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$\text{MA}_{\aleph_1}$ implies $\text{LD}(\omega_1, \omega)$

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Dichotomy for many lists, repeated

Definition

For λ an infinite regular cardinal, let $\text{LD}(\lambda^+, \lambda)$ be the assertion that for all λ -many lists $\langle \vec{L}_j := \langle L_{\delta,j} \mid \delta \in \Delta_j \rangle \mid j < \lambda \rangle$ in $\text{TL}(\lambda^+)$ either:

- there exists $A \in [\lambda^+]^{\lambda^+}$ such that for every $\delta < \lambda^+$ there exists $j < \lambda$ with $\rho_{\vec{L}_j}(A \cap \delta) = 1$, or
- there exists a sequence $\langle A_i \mid i < \lambda \rangle$ such that $\bigcup_{i < \lambda} A_i = \bigcup_{j < \lambda} \Delta_j$ and, for all $i, j < \lambda$ and $X \in [A_i \cap \Delta_j]^\lambda$ we have $\rho_{\vec{L}_j}(X) \geq \lambda$.

$\text{MA}_{\aleph_1}$ implies $\text{LD}(\omega_1, \omega)$

Assume that $\vec{L}_j := \langle L_{\delta,j} \mid \delta \in \Delta_j \rangle$ is in $\text{TL}(\omega_1)$ for every $j < \omega$.

Let condition (1) of Definition of $\text{LD}(\omega_1, \omega)$ fail: for every $A \in [\omega_1]^{\omega_1}$, there is $\delta < \omega_1$ such that $\rho_{\vec{L}_j}(A \cap \delta) > 1$ for every $j < \omega$.

We define a forcing notion $\mathbb{A}_{\vec{L}}$ where a condition p is as follows:

- p is a finite function with domain a subset of ω ;
- for $n \in \text{dom}(p)$, $p(n) = (A_n^p, B_n^p)$ where $A_n^p, B_n^p \in [\bigcup_{j < \omega} \Delta_j]^{<\omega}$.

We define the order by $q \leq p$ iff:

- $\text{dom}(q) \supseteq \text{dom}(p)$ and for $n \in \text{dom}(p)$ and $j \leq \max(\text{dom}(p))$:
 - ▶ $A_n^q \supseteq A_n^p$,
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$\mathbb{A}_{\vec{L}}$ is strong Y -c.c.

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For $j < \omega$, uncountable subsets $R, T \subseteq \omega_1$, and two sequences of distinct elements $\langle a_\alpha \mid \alpha \in R \rangle \subseteq \Delta_j$ and $\langle b_\beta \mid \beta \in T \rangle \subseteq \Delta_j$, there are $R' \in [R]^{\omega_1}$ and $T' \in [T]^{\omega_1}$ such that $a_\alpha \notin L_{b_\beta, j}$ for every $\alpha \in R'$ and $\beta \in T'$.

Proof: Consider the set $Z(R) := \{\beta < \omega_1 \mid |\{\alpha \in R \mid \beta \in L_{a_\alpha, j}\}| = \aleph_1\}$. To see it is uncountable, fix $\beta < \omega_1$. As $C(\vec{L}_j) = \{\gamma \in \text{acc}(\kappa) \mid \forall \delta \in \Delta \setminus \gamma \ (\sup(L_{\delta, j} \cap \gamma) = \gamma)\}$, let $\gamma := \min(C(\vec{L}_j) \setminus (\beta + 1))$ and $\eta := \sup\{\alpha \in R \mid a_\alpha \leq \gamma\}$. As $\gamma \in C(\vec{L}_j)$, for $\alpha \in R \setminus (\eta + 1)$ we have $\sup(L_{a_\alpha, j} \cap \gamma) = \gamma$. Hence there is $c_\alpha \in (\beta, \gamma)$ such that $c_\alpha \in L_{a_\alpha, j}$. So there is $c \in (\beta, \gamma)$ so that $\{\alpha \in R \mid c = c_\alpha \in L_{a_\alpha, j}\}$ is uncountable.

As $Z(R)$ is uncountable and (1) fails, there is $\delta < \omega_1$ so that $\rho_{\vec{L}_j}(Z(R) \cap \delta) > 1$. We claim that for $d \in Z(R) \cap \delta$, the set $\{\beta \in T \mid d \notin L_{b_\beta, j}\}$ is uncountable. Otherwise for all $d \in Z(R) \cap \delta$ the set $\{\beta \in T \mid d \notin L_{b_\beta, j}\}$ is countable. As $Z(R) \cap \delta$ is countable and T is uncountable, for some $\beta \in T$, for all $d \in Z(R) \cap \delta$ we have $d \in L_{b_\beta, j}$ which contradicts the assumption that $\rho_{\vec{L}_j}(Z(R) \cap \delta) > 1$.

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Weakly compact cardinals

Definition

An uncountable cardinal κ is *weakly compact* if for every function $f : [\kappa]^2 \rightarrow 2$ there is a homogeneous set of size κ , i.e. $\kappa \rightarrow (\kappa)_2^2$.

κ is weakly compact iff any collection of κ -satisfiable $L_{\kappa\kappa}$ -sentences, using at most κ non-logical symbols, is satisfiable.

κ is weakly compact iff it is inaccessible and every κ -tree has a branch.

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Consistency at ω_2

For an ordinal α , V_α is the set all well-founded sets of rank smaller than α .

For a set M , a formula φ is Π_1^1 over M if it is of the form $(\forall X)\psi$ where X is a second order variable (interpreted as a subset of M) and ψ may have subsets and elements of M as parameters.

For a regular uncountable cardinal κ , a filter $\mathcal{F}$ over κ is *normal* if it does not contain any bounded subset of κ and for every regressive function $f : S \rightarrow \kappa$ for $S \in \mathcal{F}$, there is $\alpha < \kappa$ such that $f^{-1}[\{\alpha\}] \in \mathcal{F}$.

κ is *weakly compact* if for every Π_1^1 formula φ over V_κ and every $R \subseteq V_\kappa$, if $(V_\kappa, \in, R) \models \varphi$, then there is an $\alpha < \kappa$ such that $(V_\alpha, \in, R \cap V_\alpha) \models \varphi$.

Then the family $\{U_{R,\varphi} \mid R \subseteq V_\kappa \text{ and } \varphi \text{ is a } \Pi_1^1\text{-formula over } V_\kappa\}$, where $U_{R,\varphi} := \{\alpha < \kappa \mid \text{if } (V_\kappa, \in, R) \models \varphi, \text{ then } (V_\alpha, \in, R \cap V_\alpha) \models \varphi\}$, generates a normal and $< \kappa$ -complete filter on κ (denoted $\mathcal{U}$). Then

- The set of all strongly inaccessible cardinals below κ is in $\mathcal{U}$;
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Consistency at ω_2 , cont'd

Theorem

Suppose that κ is a weakly compact cardinal and $2^\kappa = \kappa^+$. Then there is a partial order $\mathbb{P}$ such that both CH and $\text{LD}(\omega_2, \omega_1)$ hold in the forcing extension by $\mathbb{P}$.

The proof uses a modification of the method developed by Laver and Shelah for a proof of the consistency of SH_{ω_2} and CH.

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The poset

Let $\vec{L} := \langle \vec{L}_j := \langle L_{\delta,j} \mid \delta \in \Delta_j \rangle \mid j < \omega_1 \rangle$ be a sequence of ω_1 -many transitive ω_2 -lists. We define forcing $\mathbb{A}_{\vec{L}}$ where a condition p is as follows:

- p is a partial map with countable domain which is a subset of ω_1 ;
- for $i \in \text{dom}(p)$, $p(i) = (A_i^p, B_i^p)$ for $A_i^p \in [\bigcup_{j \leq i} \Delta_j]^{\leq \omega}$, $B_i^p \in [\omega_2]^{\leq \omega}$.

The order is defined by letting $q \leq p$ iff:

- $\text{dom}(q) \supseteq \text{dom}(p)$ and for every $i \in \text{dom}(p)$ and $j < \text{ssup}(\text{dom}(p))$:
 - ▶ $A_i^q \supseteq A_i^p$ and $B_i^q \supseteq B_i^p$,
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Lemma

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Then $\mathbb{A}_{\vec{L}}$ is a σ -closed partial order such that in the forcing extension there exists a partition $\langle A_i \mid i < \omega_1 \rangle$ of $\bigcup_{j < \omega_1} \Delta_j$ such that for all $i, j < \omega_1$ and $X \in [A_i \cap \Delta_j]^{\omega_1}$ we have $\rho_{\vec{L}_j}(X) \geq \omega_1$.

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Iteration

Assume GCH, that κ is weakly compact, and $\mathcal{U}$ a normal, $< \kappa$ -closed filter on κ . Let $\mathcal{P}$ be the set of pairs $(\mathbb{R}, \tau)$ so that $\mathbb{R}$ is a forcing in H_{κ^+} , and τ an $\mathbb{R}$ -very nice name for an element of $\text{TL}(\kappa, \omega_1)$. Fix an enumeration $\langle (\mathbb{R}_\xi, \tau_\xi) \mid \xi < \kappa^+ \rangle$ of $\mathcal{P}$ so that each pair is listed cofinally often.

We force with $\mathbb{P}_{\kappa^+}$, where $(\langle \mathbb{P}_\xi \mid \xi \leq \kappa^+ \rangle, \langle \dot{Q}_\xi \mid \xi < \kappa^+ \rangle)$ is the countable support iteration satisfying that for every $\xi < \kappa^+$:

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Theorem

If $\mathbb{P}_{\kappa^+}$ is κ -c.c., then $\mathbb{1}_{\mathbb{P}_{\kappa^+}}$ forces $2^{\aleph_0} = \aleph_1$ and $\text{LD}(\omega_2, \omega_1)$.

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 - ▶ for $\beta < \kappa$ and $j < \omega_1$, $\dot{L}_{\beta,j}^\xi$ is the lift of $\dot{L}_{\beta,j}$ from a $\mathbb{P}_\eta$ to a $\mathbb{P}_\xi$ name.
- In the remaining cases we define $\dot{Q}_\xi$ to be the trivial forcing.

Theorem

If $\mathbb{P}_{\kappa^+}$ is κ -c.c., then $\mathbb{1}_{\mathbb{P}_{\kappa^+}}$ forces $2^{\aleph_0} = \aleph_1$ and $\text{LD}(\omega_2, \omega_1)$.

Technical setup for the inductive hypothesis

We assume that for $\xi \leq \kappa^+$ the forcing $\mathbb{P}_\xi$ consists of ξ -sequences whose each element is either a map with domain subset of ω_1 and the value at each point is a pair of countable subsets of κ or the empty set (except the first element which is an element of $\text{Col}(\omega_1, \kappa)$).

For $\xi < \kappa^+$ enumerate it as $\vec{\xi} = \{\xi_\gamma \mid \gamma \in |\xi|\}$. Now fix $\xi < \kappa^+$.

Given a condition $p \in \mathbb{A}_{\vec{L}}$ and α an ordinal, we define $p|_\alpha$ to be the map with domain $\text{dom}(p) \cap \alpha$ such that for every $i \in \text{dom}(p) \cap \alpha$ we have $p|_\alpha(i) := (A_i^p \cap \alpha, B_i^p \cap \alpha)$.

Let $\mathbb{A}_{\vec{L}}|_\alpha$ be the sub-forcing defined by $p \in \mathbb{A}_{\vec{L}}$ iff $p|_\alpha = p$, where the order is defined as in $\mathbb{A}_{\vec{L}}$.

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Technical setup for the inductive hypothesis, cont'd

Let $\alpha < \kappa$, $\eta \leq \xi$, and $f \in \mathbb{P}_\eta$. Let $f|_\alpha^\xi$ be the function h with domain η such that

- If $\nu \in \{\xi_\gamma \mid \gamma < \alpha\} \cap \eta$, then
 - ▶ $\nu = 0 \implies h(\nu) = f(0) \upharpoonright (\alpha \times \omega_1)$;
 - ▶ $\nu > 0 \implies h(\nu) = f(\nu)|_\alpha$,
- otherwise $h(\nu) = \emptyset$.

Definition

Let $\alpha < \kappa$ and $\eta \leq \xi$. We define $\mathbb{P}_\eta|_\alpha^\xi = \{f \in \mathbb{P}_\eta \mid f|_\alpha^\xi = f\}$.

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Suppose that $0 < \eta \leq \xi$ and $\alpha < \kappa$. We say that $\#_\alpha^{\xi,\eta}(f, g, h)$ holds if and only if $f, g \in \mathbb{P}_\eta$ and $h \in \mathbb{P}_\eta|_\alpha^\xi$ is such that $f|_\alpha^\xi = g|_\alpha^\xi = h$.

For simplification, we write $\#_\alpha^{\xi,\eta}(f, h)$ to denote $\#_\alpha^{\xi,\eta}(f, f, h)$ and $\#_\alpha^\xi(f, g, h)$ to denote $\#_\alpha^{\xi,\xi}(f, g, h)$.

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Inductive hypothesis

In order to prove that $\mathbb{P}_\xi$ satisfies the κ -c.c. for every $\xi \leq \kappa^+$, we will need the following inductive hypothesis.

Inductive hypothesis: For all $\eta < \xi$, the poset $\mathbb{P}_\eta$ satisfies the following property, called the $\mathcal{U}^+$ -cross condition: For all $W \in \mathcal{U}^+$ and all sequences $\langle \langle f_\alpha, g_\alpha, h_\alpha \rangle \mid \alpha \in W \rangle$ such that $\#_\alpha^\eta(f_\alpha, g_\alpha, h_\alpha)$ for every $\alpha \in W$, there exists a $U \in \mathcal{U}^+$ subset of W such that for every $\alpha < \beta$ in U conditions f_α and g_β are $\mathbb{P}_\eta$ -compatible.

Lemma

Suppose $\mathbb{P}_\eta$ satisfies the $\mathcal{U}^+$ -cross condition. Then $\mathbb{P}_\eta$ satisfies the $\mathcal{U}^+$ -Knaster condition and in particular the κ -chain condition.

Proof: We fix a sequence of conditions $\langle p_\alpha \mid \alpha < \kappa \rangle$ in $\mathbb{P}_\eta$. Since $\kappa \in \mathcal{U}^+$ and $\#_\alpha^\eta(p_\alpha, p_\alpha, p_\alpha \restriction \alpha)$ trivially holds for all $\alpha < \kappa$. By the assumption there exists $U \in \mathcal{U}^+$ such that for every $\alpha < \beta$ in U the conditions p_α and p_β are $\mathbb{P}_\eta$ -compatible which is as sought.

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